

Question & Answer Defence Pack

Everything behind the notebook and the deck: the numbers, the choices we made, why we made them, and the honest answer to every question a panel can ask. Each entry gives the short answer to say out loud, then the detail to fall back on if they push.

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How to use this document. Sections 1 to 9 follow the order of the notebook, so a question about any slide lands in the matching section. Section 10 collects the hostile questions, the ones designed to find a crack. Section 11 says who answers what. The appendices hold every number we quote and a plain-language glossary, so nobody has to reconstruct a figure under pressure.

Three rules for the room. First, say the caveat before they find it: the money is synthetic, the losses are 2.18% of deals, nothing here is causal. Second, never invent a number; if you do not have it, say "that is in the notebook, I would rather check it than guess". Third, when a question goes deeper than your slide, hand it out loud to the member who built that part.

0 The project in sixty seconds

DERCO Chile gave us 550,033 retail vehicle sales from May 2009 to March 2022 and one business question: after 14 years, where is the retail model leaking value, and what should change to grow margin. We ran the full BI pipeline in one reproducible notebook: frame the decision, prepare and audit the data, build two models, communicate a recommendation, and state the ethical limits. The answer is one decision with three funded moves: retain the customers who come back, formalise the Chinese-brand portfolio, and plug the margin leak with a pre-approval check on the riskiest deals.

51.8%

of 2022 sales are Chinese brands, up from 12.2% in 2009

85.9%

of customers buy exactly once in 14 years

25.7%

of margin comes from the 14.0% Champions

2.18%

of deals close below cost, about 6.8 bn CLP

38.8%

of volume is Suzuki alone, a concentration risk

11×

more below-cost deals in own stores than at dealers

50.6%

of loss-making deals caught by reviewing the riskiest 10%

0.83

cross-validated ROC-AUC of the classifier

1 Framing, scope and the client

Q What exactly did the client ask for, and who is the decision-maker?

A The Commercial Director. They own brand mix, the dealer-versus-own-store channel, and the marketing budget, so they are the only person who can act on all three of our moves. The question they asked is where the retail model leaks value and what to change to grow margin.

Say: "We were not asked for a model. We were asked where the money leaks and what to change, and every deliverable is built backwards from that."

Q Why those six KPIs and not others?

A Transactions and revenue measure scale; retail margin and margin percentage measure what the business keeps; repeat-customer rate measures whether growth compounds; loss-making-deal rate measures leakage. Each of the three moves is tied to one of them, so a KPI that moved nothing would have been dropped.

Q How do you define success for the project itself?

A A ranked, peso-valued list of moves the Commercial Director can fund next year, not a high accuracy score. That is why the headline number for Model B is "half the leak caught by reviewing 10% of deals" rather than the AUC.

Q Why two models instead of one?

A The question has two halves. "Who should we keep" is a customer-level, unsupervised question, so it gets RFM and KMeans. "Which deals leak margin" is a deal-level, supervised question with a labelled target, so it gets a classifier. One model cannot answer both without pretending the two questions are the same.

2 The data, the audit and the cleaning decisions

Q What does one row represent, and how big is the file?

A One row is one car sale. 550,033 raw rows, 550,032 after cleaning, 464,226 unique customers, 9 brands, 1,543 model-and-trim strings, 511 comunas, from 5 May 2009 to 28 March 2022. The source is a single Excel sheet whose one column packs all ten fields pipe-delimited, so the first step is to split it and assign real types.

Q Can this data be trusted?

A Yes, and we checked rather than assumed. Zero exact duplicate rows. Zero rows where the date's year disagrees with the `year` column, which is why we drop that column as redundant. Missing values are about 0.72% and confined to `direccion` and `comuna`. One row has an unparseable list price. The only large anomaly is a real business finding, not a data error: 11,966 sales with negative margin.

Q Why fill missing comunas with UNKNOWN instead of dropping those rows?

A Because those rows carry a valid brand, price and margin. Dropping them would quietly reduce every revenue and margin total, which hides money. Marking geography as unknown loses one attribute; dropping the row loses the sale.

Say: "We delete data only when the row cannot answer any question. One row met that bar."

Q Negative margins look like outliers. Why keep them?

A They are the finding. 11,966 deals, 2.18% of the file, roughly 6.8 bn CLP of leaked margin, about 2.7% of all retail margin. If we treated them as outliers and removed them, we would have deleted the single most actionable result in the project and Model B would have had no target left to predict.

Q What did you engineer, and why each one?

A `brand_origin` (Chinese versus Incumbent) because the strategic question is about origin, not about nine individual brands. `margin_pct` because absolute margin confounds price with pricing health. `loss_making` as the model target. `price_band` as a tercile of list price (Economy, Mid, Premium) so price can enter descriptive views as a category. And year, month, quarter rebuilt from the date for the time views.

Q Which brands count as Chinese, and who decided?

A JAC Cars, Great Wall, Changan, Geely and Haval. The rest — Suzuki, Mazda, Renault, Samsung — are incumbents. It is an explicit set in the notebook, not an inferred rule, so anyone can challenge or edit it in one line.

Q Why Parquet rather than CSV for the clean table?

A Parquet keeps column types, so a date stays a date and the category stays a category on reload. It is also less than half the size: 51 MB against 109 MB as CSV. We ship a 2,000-row CSV sample for anyone who wants to look at the data in a spreadsheet, and the README shows the one line that writes a full CSV.

3 What the exploratory analysis showed

The descriptive findings, with the exact figures behind each claim

FINDING	THE NUMBERS
Brand concentration	Suzuki 38.8%, Mazda 17.8%, Renault 10.9%, JAC 9.1%, Great Wall 8.5%, Changan 8.2%, Samsung 3.5%, Geely 1.7%, Haval 1.6%
The structural shift	Chinese-brand share 12.2% (2009) to 51.8% (2022); rose to 31.1% in 2012, fell to 19.5% in 2016, climbed again from 2019
Channel mix and pricing	ces 357,983 deals and 6.24% margin; propio 192,049 deals and 5.20% margin, a gap of 1.04 points
Where the leak sits	propio loses money on 5.38% of deals against 0.46% at ces, about 11 times more often
Loss rate by brand	Mazda 4.15% (-3.42 bn), Samsung 3.65%, Haval 3.43%, Renault 2.18%, Suzuki 1.78% (-1.61 bn), Changan lowest at 1.05%
Geographic concentration	top 10 comunas 33.4% of sales, top 20 49.8%, top 50 74.6%, out of 511 comunas
Seasonality	August peaks at 10.6% of annual volume, December 10.3%, April troughs at 6.9%

Q Chinese brands went from 12% to 52%. Is that not just a partial-year artefact?

A No, and we say the caveat before the chart. 2009 starts in May and 2022 ends in March, so both end years are partial. That is exactly why we compare *shares* and never totals at the two ends. A share is unaffected by how many months the year contains. The path also supports the claim: it is not a straight line, it dips to 19.5% in 2016 and recovers from 2019, and it crosses 50% only in the final year.

Q Why is Mazda's leaked amount the largest if its loss rate is not the highest by much?

A Because leakage is rate times volume times ticket size. Mazda is 4.15% of 97,745 deals at a high average price, so it accounts for 3.42 bn of the 6.84 bn. Suzuki has a much lower rate, 1.78%, but on 213,402 deals, which still adds 1.61 bn. That is why the recommendation targets deals, not brands.

Q Why does the dealer channel price better than DERC's own stores?

A We do not know, and we say so. Plausible drivers are discount authority sitting lower in an own store, or own stores absorbing trade-ins and fleet deals that dealers pass on. We have no field in this dataset that separates those, which is precisely why we recommend *reviewing* those deals rather than closing or restructuring the channel.

Say: "That is correlation. The honest response is a review process, not a reorganisation."

4 Model A — customer segmentation

Q How is the RFM table built?

A One row per customer hash. Recency is days since the last purchase, measured against a reference date of the last date in the file plus one day. Frequency is the number of purchases. Monetary is total list price. We also carry total margin per customer so we can report margin share by segment.

Q Why log-transform and standardise before clustering?

A KMeans measures Euclidean distance, so a skewed axis dominates the geometry. Frequency and monetary are heavily right-skewed because a handful of fleet buyers spend orders of magnitude more than the median, so we apply `log1p` to compress that tail, then standardise all three axes to mean 0 and standard deviation 1 so no single variable outweighs the others. We compress the outliers rather than delete them, because fleet buyers are real customers.

Q How did you choose $k = 4$? The silhouette says otherwise.

A We tested $k = 2$ through 7 with the elbow and the silhouette, and we state the trade-off instead of hiding it. $k = 2$ scores highest at 0.539 but only separates recent from old customers, which is one bit of information and no marketing action. $k = 5$ scores 0.400 against 0.393 for $k = 4$, a difference of 0.007 that is noise, and it splits a group the CRM would treat identically. $k = 4$ sits at the elbow, holds a usable silhouette, and maps one-to-one onto the four plays a CRM actually runs: retain, grow, nurture, cap spend.

Say: "Silhouette alone does not settle k . We broke the tie on actionability and we published the whole curve so you can disagree with us."

Silhouette by k , on a seeded 30,000-customer sample for speed

K	2	3	4	5	6	7
silhouette	0.539	0.468	0.393	0.400	0.391	0.384

The four segments, named from the cluster profile by descending monetary value

SEGMENT	CUSTOMERS	MARGIN	AVG BUYS	PLAY
Champions (repeat, high-value)	14.0%	25.7%	2.31	Retain
Big-ticket one-timers	22.8%	30.6%	1.00	Grow
Mainstream one-timers	32.5%	24.3%	1.00	Nurture
Dormant / lapsed	30.8%	19.5%	1.00	Cap spend

Q Big-ticket one-timers carry more margin than Champions. Why is the recommendation about Champions?

A Because a retention program needs customers who can come back, and Champions are the only group that ever does, at 2.31 purchases on average. Big-ticket one-timers hold 30.6% of margin but buy once, so the play there is different: grow the basket at the point of sale, not run loyalty spend against people who never return. Both plays are in the table; we lead with retention because it is the cheaper margin.

Q Are the segment names not just labels you invented?

A The names are ours; the groups are not. Clusters are ordered by mean monetary value and named from the profile table, which shows recency, frequency, monetary and margin share per cluster. Anyone can rename them; the partition and the numbers do not change.

Q Why KMeans instead of RFM quantile scoring, hierarchical clustering or DBSCAN?

A Quantile scoring would impose the segments rather than discover them, which defeats the purpose of the exercise. Hierarchical clustering does not scale comfortably to 464,226 customers. DBSCAN needs a density structure this data does not have, and would push most one-time buyers into noise. KMeans is the standard, it scales, and it is inspectable, which matters more here than squeezing out a marginally better partition.

5 Model B — predicting loss-making deals

Q What is the target and what does the model see?

A The target is `loss_making`, one when margin is below zero. The features are only what is known before a deal closes: list price, month and year as numeric; brand, channel and comuna as categorical, with comuna limited to the 30 largest and the tail bucketed as OTHER to control cardinality. Numeric features are standardised and categoricals one-hot encoded inside a pipeline, so the preprocessing is fitted on the training fold only.

Q Where is the leakage risk, and how did you avoid it?

A This is the most important choice in the whole model. The target is derived from `margen_retail`, so feeding the model `margen_retail` or `margin_pct` would let it read the answer instead of predicting it. Both are excluded, along with anything else derived from margin. The resulting scores are lower than a leaky model would show, and they are real.

Say: "If we had left margin in, the AUC would have been near 1.0 and the model would have been worthless in production, because at quotation time nobody knows the margin yet."

Q How was the model validated?

A A stratified 75/25 split, 412,524 training rows and 137,508 test rows, with stratification so the 2.18% base rate holds in both. Then 5-fold cross-validation on the full dataset. We also ran two baselines first, because a score means nothing without one.

Model comparison on the hold-out test set. Base rate, and therefore random PR-AUC, is 0.022

MODEL	ROC-AUC	PR-AUC	READ
Dummy (stratified)	0.501	0.022	the floor, equals random
Logistic regression, balanced	0.813	0.077	a linear model already beats random
HistGradientBoosting, balanced	0.855	0.142	chosen; 5-fold CV 0.828, std 0.016

Q Why do you quote 0.83 when the hold-out says 0.855?

A Because the 5-fold folds run 0.808 to 0.849 with a mean of 0.828 and a standard deviation of 0.016. The single hold-out split sits at the top of that range, so quoting it would be quoting our luckiest number. The cross-validated mean is the honest generalisation estimate.

Q Why ROC-AUC and PR-AUC rather than accuracy?

A Accuracy is meaningless at a 2.18% base rate: predicting "never a loss" scores 97.8% and finds nothing. ROC-AUC measures ranking quality across all thresholds, which is what a review queue needs. PR-AUC is the honest view under imbalance because it ignores the enormous true-negative mass; at 0.142 against a 0.022 base rate, it is about 6.5 times better than random.

Q How does the score become a business decision?

A We do not use a 0.5 threshold. We rank the test deals by predicted risk and take as many as the pricing team can actually review. At 10% of deals, the queue contains 50.6% of all loss-making deals, 11.0% of the flagged deals truly lose money, and that is 5.1 times better than reviewing a random 10%. The threshold is a capacity decision the client sets, not a statistic we impose.

Say: "The deliverable is a daily watchlist sized to the team that has to work it."

Q What drives the risk?

A Permutation importance, measured as the drop in ROC-AUC when a feature is shuffled, on a seeded 20,000-row sample with 5 repeats: brand, list price and channel lead. That is consistent with the EDA, where Mazda and Samsung carry the highest loss rates and own stores close below cost about 11 times more often than dealers. Two independent views agreeing is a good sign; neither of them is a causal claim.

Q Why HistGradientBoosting rather than XGBoost, a random forest or a neural network?

A It is in scikit-learn, so it needs no extra dependency; it handles mixed tabular features and interactions natively; it trains on 550,000 rows in seconds; and it supports `class_weight='balanced'`, which matters at a 2.18% base rate. On tabular data of this size a neural network would add complexity and training time without a credible accuracy gain. We also report a logistic baseline, so the panel can see exactly what the extra model complexity bought: 0.813 to 0.855 ROC-AUC and 0.077 to 0.142 PR-AUC.

Q Did you tune hyperparameters?

A No. We used the library defaults with balanced class weights and spent the effort on the things that actually decide whether the model is usable: excluding leakage, stratifying the split, cross-validating, comparing against two baselines, and converting the score into a capacity-sized review rule. A tuning sweep would likely add a small amount of AUC and would not change any recommendation. If the client deploys this, tuning and calibration belong in that project, along with a monitored flag rate.

6 The recommendation and its value

Q What are you actually asking the client to do?

A Three funded moves for FY2026. Retain: a loyalty program aimed at Champions and at converting one-timers. Formalise China: treat Chinese brands as a first-class portfolio in pricing, stock and marketing, because they are already the majority of sales. Plug the leak: run the classifier as a pre-approval check on the riskiest 10% of deals, starting with the own-store channel where the leak concentrates.

Q How much money does this recover?

A In relative terms, the leak is about 2.7% of all retail margin, and reviewing the riskiest 10% of deals puts about half of it in front of a human. We deliberately do not promise a peso figure, because the money fields are synthetic. The right way to size it is to run the review for one quarter and measure the margin saved on flagged deals against the review cost.

Q What would you measure to know the recommendation worked?

A Loss-making-deal rate overall and by channel, which should fall; repeat-customer rate, which should rise; margin percentage by channel; and the flag rate by channel and comuna, which should be monitored so the review does not simply harden a bias. Those are the same KPIs we defined in Stage 1, which is the point of defining them before the analysis.

Q Why not recommend dropping Suzuki or closing own stores?

A Because the evidence does not support either. Suzuki is 38.8% of volume and its loss rate is below average at 1.78%; concentration is a risk to manage, not a brand to cut. Own stores are 35% of volume and the loss is concentrated in a subset of their deals, which is an argument for a pricing control, not for closing a channel. We recommend the smallest intervention the evidence justifies.

7 Ethics, privacy and bias

Q The customer ID is hashed. Is the data anonymous?

A No, and treating a hash as anonymity is a common mistake. A RUT is short and check-digit validated, so anyone with the same hash function can enumerate candidate RUTs and match them back. We treat the column as personal data: it is only a grouping key for the RFM table, never a model feature, and it never appears at row level in the deck, the PDF or the dashboard.

Q What did you do about the address field?

A Dropped it in Stage 2 as data minimisation. It is not in the clean Parquet, not in the sample CSV and not in any model. We kept comuna because municipality is the geographic unit the decision needs and it is coarse enough not to locate a household.

Q Where is this analysis biased?

A Four places, and we name them before anyone asks. Geographically: the top 10 comunas hold 33.4% of rows, so "valuable customer" is partly a proxy for "urban customer", and a retention budget allocated straight off the segment table would under-serve rural comunas. Survivorship: we only see closed DERCO sales, never the customers who walked away or bought a competitor. Feedback loop: a deployed Model B flags own-store deals more often, so reviewers see more of them and confirm the pattern, which is why flag rate by channel and comuna has to be monitored and the model refitted on post-deployment data. And segment realism: a silhouette of 0.39 means the four groups are a marketing convenience, not four natural populations, so no customer should ever be told which one they are in.

Q Would you let this model reject a deal automatically?

A No. Of the deals it flags, 11.0% actually lose money, which is excellent for triage and unacceptable for automatic rejection. It produces a review queue; a human decides. That is stated in the notebook, on the slide, and in the recommendation itself.

8 Process, tooling and reproducibility

Q How do we know the numbers on the slides match the notebook?

A Because no number on a slide is typed by hand. Stage 4 writes `presentation/metrics.json`, and both the HTML deck and the PowerPoint builder read from it. If the analysis changes, the slides change on the next build or they break loudly. The same file feeds the static dashboard image.

Q Is the analysis reproducible?

A Yes. One seed, `SEED = 42`, governs the KMeans initialisation, the silhouette sample, the train/test split, the cross-validation folds, the permutation-importance sample and the CSV sample. A full Restart and Run All takes about five to ten minutes and regenerates the clean Parquet, the CSV sample, the dashboard image and the metrics file identically. We verified this: re-running produced byte-identical `metrics.json`, `dashboard.png` and sample CSV.

Q Can the notebook run somewhere other than your laptop?

A Yes. Paths are resolved at run time rather than hard-coded: the notebook walks up from the working directory looking for a `data/` folder or the source workbook, so it runs unchanged from `notebooks/`, from the repository root, from the flat submission folder, or from Colab once the workbook is uploaded. It prints the folders it resolved in the setup cell.

Q Why a notebook rather than Power BI or Tableau?

A Because the deliverable has to be auditable end to end. A notebook shows the cleaning decisions, the model choices and the validation next to their outputs, which a dashboard tool hides. We still deliver the dashboard: the six-KPI view exists as an image and as an interactive HTML page, both fed by the same metrics file.

Q How was AI used, and what is yours?

A An AI coding assistant was used as a pair-programmer: boilerplate such as plot styling and pipeline wiring, structural suggestions for the RFM and classifier code, and drafts of explanatory prose. The business framing, the methodological choices — $k = 4$, the leakage rule, the 10% review threshold — the interpretation and the recommendation are ours, every line was reviewed, and no number was written by hand: everything reproduces on a Restart and Run All. The disclosure is in notebook section 6 and in the README.

9 The hard questions

Q Your money is synthetic. Why should the client believe any of this?

A Because the obfuscation preserves the structure we use. Shares, rankings and trends are computed from ratios of the same transformed field, so "Chinese brands are the majority", "2.18% of deals close below cost" and "own stores lose money 11 times more often" hold regardless of the scaling. What does not hold is the absolute level, which is why we never present 6.8 bn CLP as an audited number and why the first action we recommend is a one-quarter measurement against real financials.

Q PR-AUC of 0.14 is a weak model.

A In absolute terms it is modest and we say so on the slide. In context it is 6.5 times the 0.022 base rate, and the decision it supports does not need precision: it needs a ranking good enough that a fixed review capacity finds most of the losses. At 10% capacity it finds 50.6%. If we needed to auto-reject deals, this model would not be good enough, and that is exactly why we do not propose that.

Q Your segmentation has a silhouette of 0.39. That is weak clustering.

A Agreed, and we published the full curve rather than the best-looking number. The data is dominated by one-time buyers, so there are no cleanly separated spheres to find; any method would show this. The segments are still useful because the business action differs by group and the margin shares differ materially. We treat them as a marketing partition to be revalidated yearly, not as a discovered truth.

Q Champions have more margin because they buy more. Is that not circular?

A Partly, and it is worth separating. The circular part is that frequency feeds both the cluster and the margin total. The non-circular part is the ratio: 14.0% of customers carry 25.7% of margin, and they are the only group whose average purchase count exceeds one, at 2.31. The recommendation does not claim that loyalty causes margin; it claims that the only repeat behaviour in the file sits in one identifiable group, so that is where a retention test should start.

Q Is 86% buying once not just customers changing their RUT or buying through a company?

A That is the most plausible challenge to our loyalty conclusion and we flag it as a limitation. We cannot distinguish a genuine one-time buyer from an unstable key across 14 years without business confirmation. That is why the recommendation is to fund a retention test and validate the key with the CRM team first, rather than committing the full budget on our number.

Q Chinese brands grew. Does that mean DERCO should push them harder?

A It means the organisation should treat them as core rather than as a side bet, which is a structural claim about pricing, stock and marketing ownership. Whether pushing them harder increases margin is a different question that this data cannot answer: we see what was sold, not what was offered, discounted or refused. A pricing experiment would settle it.

Q You had 550,000 rows and 14 years. Why is there no forecast?

A Because the decision on the table is where to move next year's budget, not what next month's volume will be. A forecast would look sophisticated and change nothing about the three moves. If the client wants capacity planning, monthly volume with the August peak and April trough is the natural next project, and the clean table already supports it.

Q What would you do differently with more time?

A Three things. Validate the customer key with the CRM to firm up the loyalty conclusion. Calibrate Model B's probabilities and tune it, so the review threshold can be set on expected value rather than on capacity alone. And bring in one external field, either list-price history or a competitor volume index, so at least one claim could move from correlation toward causation.

10 Who answers what

Presenting order and question ownership differ on purpose. You present the slide; whoever built that part answers the deep question. Hand over out loud, by name.

OWNER	PRESENTS	ANSWERS ANYTHING ABOUT
Baptiste Vial	Slides 1 to 3	framing, scope, the client, insight synthesis, ethics and limits, the dashboard and deck build
Luis-Felipe Cáceres	Slides 4 to 6	both models: RFM construction, choice of k, leakage, validation, metrics, the 10% rule
Agustín Reyes	Slides 7 to 9	data quality, cleaning decisions, feature engineering, every EDA chart
Vicente Rodríguez	Slides 10 to 13	KPIs and success metrics, reproducibility, environment, repository and data documentation

If a question has no good answer. Say what you do know, say what would settle it, and stop. "We cannot answer that with this dataset. What would settle it is X." A clean boundary reads as competence; a guess that unravels does not.

A Appendix — every number we quote

Scope and data		Commercial findings	
Raw rows	550,033	Chinese share 2009 / 2022	12.2% / 51.8%
Rows after cleaning	550,032	Chinese share low point (2016)	19.5%
Rows dropped	1 (null list price)	Suzuki share of volume	38.8%
Unique customers	464,226	ces volume / margin	65.1% / 6.24%
Brands / models / comunas	9 / 1,543 / 511	propio margin	5.20%
Date range	2009-05-05 to 2022-03-28	Loss-making deals	11,966 (2.18%)
Missing comuna	0.72%	Leaked margin	-6.84 bn CLP
Duplicate rows	0	Share of all retail margin	about 2.7%
Date-versus-year mismatches	0	Loss rate propio / ces	5.38% / 0.46%
Zero-margin sales	26	Top-10 comuna share	33.4%
Clean table	550,032 × 15	Peak / trough month	August / April

Model A		Model B	
Customers clustered	464,226	Target base rate	2.18%
Features	R, F, M	Train / test rows	412,524 / 137,508
Transform	log1p F and M, then standardise	Split	75/25, stratified
k tested	2 to 7	ROC-AUC hold-out	0.855
k chosen	4	ROC-AUC 5-fold	0.828, std 0.016
Silhouette at k = 4	0.393	Fold range	0.808 to 0.849
Champions	14.0% cust / 25.7% margin	PR-AUC vs base	0.142 vs 0.022
Champions avg purchases	2.31	Top-10% capture	50.6%
Customers buying once	85.9%	Precision in flagged set	11.0%
		Lift over random	5.1×

B Appendix — notebook map

SECTION	WHAT IT ESTABLISHES
1.1 to 1.4	business question, decision-maker, six KPIs, project success metric, synthetic-money caveat
2.2 to 2.3	parsing the pipe-delimited sheet, typing, data dictionary
2.4 to 2.5	quality audit and every cleaning decision with its rationale
2.6	feature engineering: brand_origin, margin_pct, loss_making, price_band, calendar parts
2.7 (a) to (h)	EDA: brand mix, channel, 14-year trend, seasonality, the Chinese shift, geography, margin health, takeaways
2.8	writes the clean Parquet and the CSV sample
3A.1 to 3A.3	RFM table, choice of k with elbow and silhouette, segment naming and profile
3B.1 to 3B.6	leakage rule, baselines, gradient boosting, cross-validation, ROC and PR curves, the 10% rule, permutation importance
4.1 to 4.3	five ranked insights, the recommendation, the dashboard and the metrics export
5.1 to 5.3	privacy, bias, honest limitations
6	AI-use disclosure and contribution summary

C Appendix — glossary

TERM	IN ONE SENTENCE
RFM	Describing a customer by how recently they bought, how often, and how much they spent.
KMeans	An algorithm that groups points into k clusters by repeatedly assigning each point to the nearest cluster centre.
Elbow method	Plotting within-cluster spread against k and choosing the point where adding clusters stops helping much.
Silhouette	A score from -1 to 1 for how much closer a point is to its own cluster than to the next nearest one.
Data leakage	Letting a model see information that would not exist at prediction time, usually something derived from the answer.
Class imbalance	One outcome being far rarer than the other, here 2.18%, which makes accuracy useless as a measure.
Stratified split	Splitting train and test so the rare class keeps the same proportion in both.
Cross-validation	Rotating which fifth of the data is held out, five times, to check the score is not a lucky split.
ROC-AUC	The chance the model ranks a random loss-making deal above a random profitable one; 0.5 is random.
PR-AUC	Precision against recall summarised in one number; the fair measure when the positive class is rare.
Lift	How much better a targeted selection is than picking the same number of cases at random.
Permutation importance	Shuffling one feature and measuring how much the score drops, which shows what the model relies on.
One-hot encoding	Turning a category such as brand into one yes/no column per value so a model can use it.
Parquet	A compressed columnar file format that preserves column types, unlike CSV.

Every figure in this document is computed by `notebooks/BI2026_FinalProject.ipynb` and exported to `presentation/metrics.json`. Money fields are synthetic: shares, ranks and trends are valid, absolute CLP amounts are not. Regenerate this pack with `chrome --headless --print-to-pdf` over `presentation/qa_defense.html`.